

PIETARSAARI KALLAN, Finland

WMO: 029200

Lat: 63.751N

Lon: 22.523E

Elev: 63

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.1	1.4	-9.7	3.3	-4.9	-2.4	4.8	3.3	42.4	30.5	39.0	31.8	9.1	110	0.879

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	72.4	64.6	69.3	62.8	66.7	61.3	66.5	71.2	64.1	68.3	62.1	66.0	11.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	91.4	69.5	62.3	84.1	66.8	60.1	77.8	64.8	31.2	71.4	29.4	68.4	27.9	66.3	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.3	33.3	29.9	DB	-4.9	76.4	8.3	4.4	-10.9	79.6	-15.8	82.2	-20.5	84.7	-26.5	87.9	(4)
(5)			WB	-5.7	67.7	8.7	3.4	-11.9	70.1	-17.0	72.1	-21.9	73.9	-28.2	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	23.4	22.0	26.4	35.4	44.9	52.6	60.5	59.7	52.7	42.9	35.8	30.1	(6)	
	DBStd	14.88	9.95	11.00	7.88	4.73	6.15	5.33	5.61	4.69	4.55	5.45	6.22	8.57	(7)	
	HDD50	4286	824	785	732	438	183	29	1	1	25	224	426	618	(8)	
	HDD65	8927	1289	1205	1197	888	623	374	164	174	370	684	876	1083	(9)	
	CDD50	871	0	0	0	0	25	107	326	303	105	5	0	0	(10)	
	CDD65	36	0	0	0	0	0	1	24	11	0	0	0	0	(11)	
	CDH74	53	0	0	0	0	0	2	30	21	0	0	0	0	(12)	
	CDH80	5	0	0	0	0	0	0	3	2	0	0	0	0	(13)	
	WSAvg	15.5	16.5	15.5	15.4	14.2	14.3	13.7	12.8	13.3	15.9	18.0	18.1	18.1	(14)	
	PrecAvg	20.8	1.5	1.1	1.1	1.0	1.5	2.0	2.6	2.2	2.0	2.2	1.9	1.7	(15)	
(16) Precipitation	PrecMax	26.5	2.7	2.0	1.9	2.6	2.8	3.4	4.8	5.2	4.3	3.7	4.4	3.3	(16)	
	PrecMin	15.2	0.4	0.1	0.2	0.2	0.2	0.5	0.2	0.8	0.8	0.6	0.2	0.7	(17)	
	PrecStd	3.1	0.6	0.6	0.5	0.6	0.7	0.8	1.2	1.3	1.1	0.9	0.9	0.7	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	38.7	41.4	54.4	68.7	71.1	77.1	76.7	65.0	55.5	48.6	42.7	(19)	
		MCWB	38.1	36.1	36.3	44.8	55.4	59.8	67.1	65.5	59.3	52.5	46.3	40.8	(20)	
	2%	DB	37.2	36.9	38.6	48.2	62.8	66.0	73.6	71.1	62.0	53.4	46.8	41.4	(21)	
		MCWB	35.6	34.9	35.5	42.8	53.8	58.5	66.4	62.5	57.5	51.0	45.1	39.6	(22)	
	5%	DB	35.9	35.3	37.2	43.4	57.4	63.1	70.7	67.9	60.2	51.7	45.2	40.4	(23)	
		MCWB	34.7	33.9	34.8	39.3	50.6	56.9	64.7	62.0	56.5	49.4	43.3	38.7	(24)	
	10%	DB	34.5	33.9	35.9	41.1	53.2	60.0	68.2	65.8	58.7	50.2	43.5	38.9	(25)	
		MCWB	33.5	32.8	33.9	37.9	47.8	54.6	62.9	60.9	55.6	48.0	41.7	37.3	(26)	
	0.4%	WB	38.6	36.9	37.7	47.9	58.3	62.8	69.5	67.3	60.6	53.3	46.8	41.8	(27)	
		MCDB	39.1	38.2	40.3	50.4	65.7	68.5	73.7	73.7	63.1	54.7	47.9	42.4	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.2	35.3	36.4	42.5	54.4	59.8	67.9	64.6	58.8	51.2	44.9	40.2	(29)	
		MCDB	36.7	36.5	38.1	45.8	62.2	64.6	72.9	69.2	61.2	52.5	46.2	41.1	(30)	
	5%	WB	34.8	34.3	35.4	39.5	51.3	57.6	66.0	62.9	57.2	49.5	43.1	39.0	(31)	
		MCDB	35.6	35.2	37.0	42.1	57.5	62.2	70.4	66.7	59.3	51.0	44.1	40.2	(32)	
	10%	WB	33.5	33.2	34.1	38.0	48.5	55.1	63.8	61.5	56.0	47.9	41.7	37.5	(33)	
		MCDB	34.4	34.1	35.8	40.3	53.0	59.1	67.8	65.0	58.2	49.5	42.9	38.9	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	7.5	7.8	8.0	7.1	8.6	7.4	6.8	6.3	5.9	5.4	5.2	5.9	(35)	
		MCDBR	5.7	5.8	6.7	10.7	16.1	12.1	9.9	9.4	7.2	6.1	5.3	5.2	(36)	
		MCWBR	5.3	5.3	5.2	6.7	9.2	7.1	5.8	5.4	4.9	5.9	5.4	5.0	(37)	
	5% WB	MCDBR	5.5	5.4	6.5	9.1	15.4	11.2	9.0	7.4	6.4	6.2	5.4	5.1	(38)	
		MCWBR	5.2	5.0	5.3	6.6	9.8	7.2	6.2	5.5	5.1	6.0	5.5	5.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.228	0.287	0.312	0.351	0.350	0.351	0.371	0.358	0.333	0.314	0.364	0.351	(40)	
	taud		2.021	2.149	2.208	2.267	2.441	2.483	2.452	2.481	2.528	2.373	2.616	2.571	(41)	
	Ebn at Noon		114	198	245	261	274	276	266	259	243	191	105	56	(42)	
	Edh at Noon		9	20	28	33	31	30	30	27	21	16	8	5	(43)	
(44) All-Sky Solar Radiation	RadAvg		44	224	701	1207	1628	1799	1644	1254	717	267	62	18	(44)	
	RadStd		4	44	75	108	166	154	153	121	74	49	10	2	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page